

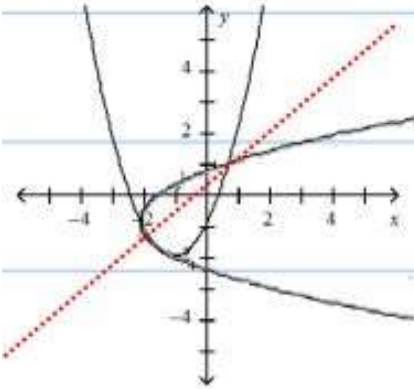
Lesson 0.1 Practice Solutions

1. For each of the following :

(a) State the domain and range

(b) Draw the inverse (on the same grid) using colored pen or pencil

(c) State the domain and range of the inverse



$$D_f: \underline{\{x | x \in \mathbb{R}\}}$$

$$R_f: \underline{\{y | y \in \mathbb{R}, y \geq -2\}}$$

$$D_f^{-1}: \underline{\{x | x \in \mathbb{R}, x \geq -2\}}$$

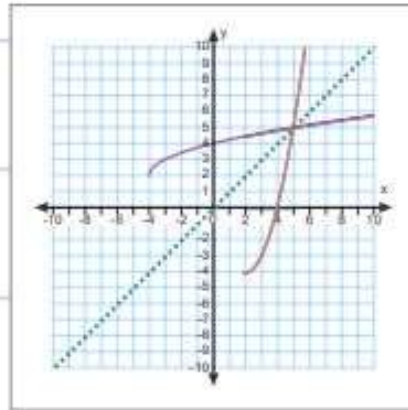
$$R_f^{-1}: \underline{\{y | y \in \mathbb{R}\}}$$

$$D_f: \underline{\{x | x \in \mathbb{R}, x \geq -4\}}$$

$$R_f: \underline{\{y | y \in \mathbb{R}, y \geq 2\}}$$

$$D_f^{-1}: \underline{\{x | x \in \mathbb{R}, x \geq 2\}}$$

$$R_f^{-1}: \underline{\{y | y \in \mathbb{R}, y \geq -4\}}$$



2. Given the two functions $f = \{(2, 4), (3, 1), (5, 0), (6, -2), (-1, 8)\}$ and

$g(x) = -\sqrt{x-1} + 1$ determine:

a) $f^{-1}(4) = \mathbf{2}$

b) $g^{-1}(x)$

c) $f^{-1}(g^{-1}(0))$

$$\mathbf{x = -\sqrt{y-1} + 1}$$

$\mathbf{f^{-1}(2) = \text{not defined}}$

$$\mathbf{(x-1)^2 = y-1}$$

$$\mathbf{y = 1 + (x-1)^2}$$

$$\mathbf{g^{-1}(x) = (x-1)^2 + 1}$$

3. Given $f(x) = x^2 + 3x - 3$ and $g(x) = f^{-1}(x)$, what are the value(s) of $g(1)$?

$$1 = x^2 + 3x - 3$$

$$x^2 + 3x - 4 = 0$$

$$(x+4)(x-1) = 0$$

$$x = -4, x = 1$$

4. Give $f(x) = -k(4 + 2x)$, determine the value of k if $f^{-1}(2) = -5$.

$$f^{-1}(2) = -5 \longrightarrow f(-5) = 2$$

$$f(x) = -k(4 + 2x) \longrightarrow 2 = -k(4 - 10)$$

$$2 = 6k$$

$$k = \frac{1}{3}$$

5. Given the diagram below, graph $2f^{-1}(x+1)$ on the same set of axes;

For $f(x)$:

$$-5 \leq x \leq 5 \text{ and } 0 \leq y \leq 5$$

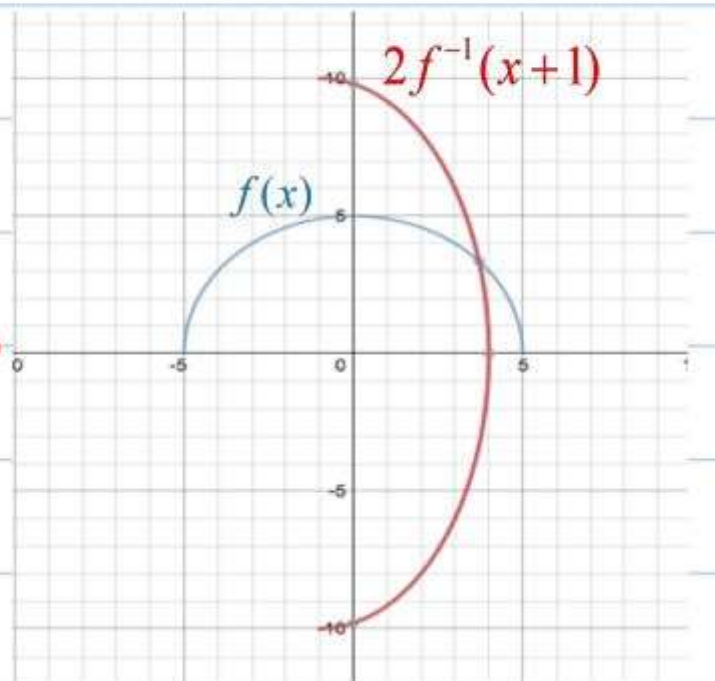
For $f^{-1}(x)$:

$$0 \leq x \leq 5 \text{ and } -5 \leq y \leq 5$$

For $2f^{-1}(x+1)$:

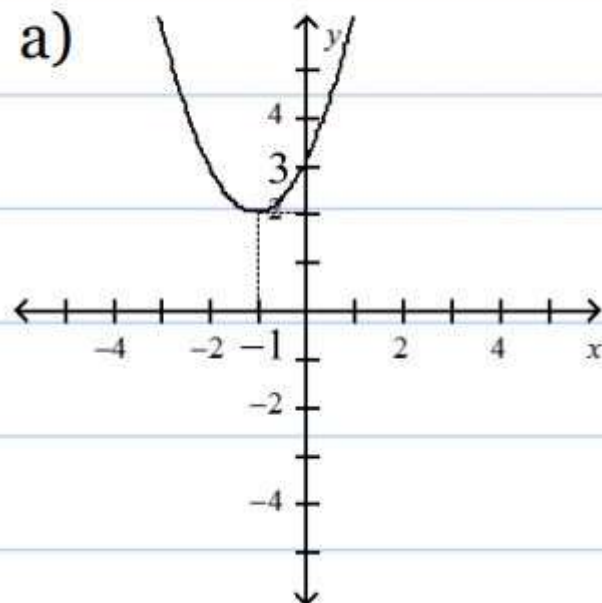
$$0 \leq x+1 \leq 5 \text{ and } -10 \leq 2y \leq 10$$

$$-1 \leq x \leq 4 \text{ and } -10 \leq 2y \leq 10$$



6. Find the equation of the following functions.

a)



$$y = a(x+1)^2 + 2$$

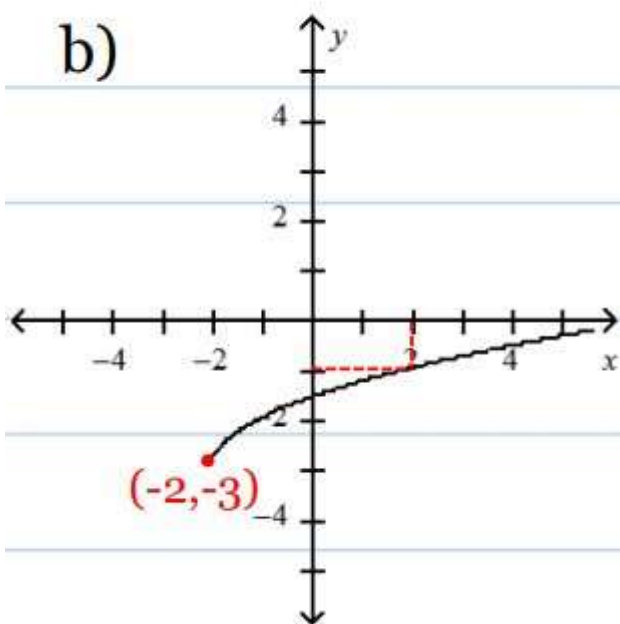
$$3 = a(0+1)^2 + 2$$

$$1 = a$$

$$\therefore y = (x+1)^2 + 2$$

7. Find the equation of the following functions.

b)



$$y = a\sqrt{x+2} - 3$$

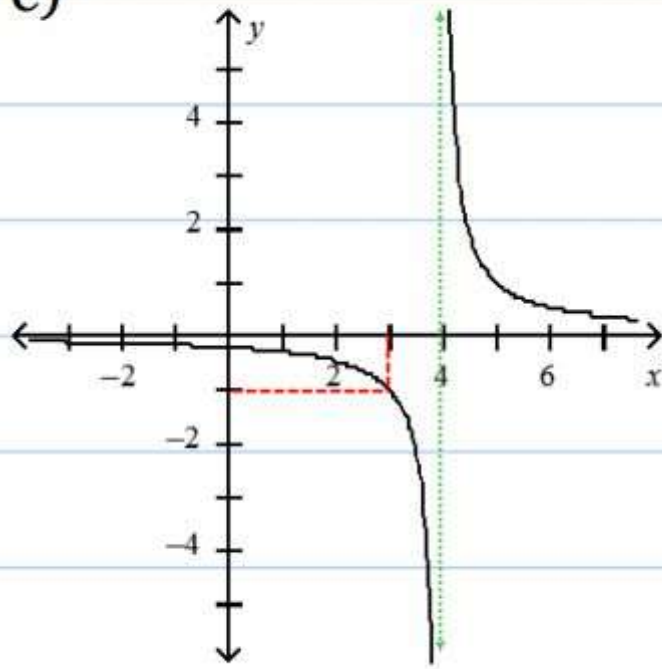
$$-1 = a\sqrt{2+2} - 3$$

$$2 = 2a$$

$$a = 1$$

$$\therefore y = \sqrt{x+2} - 3$$

c)



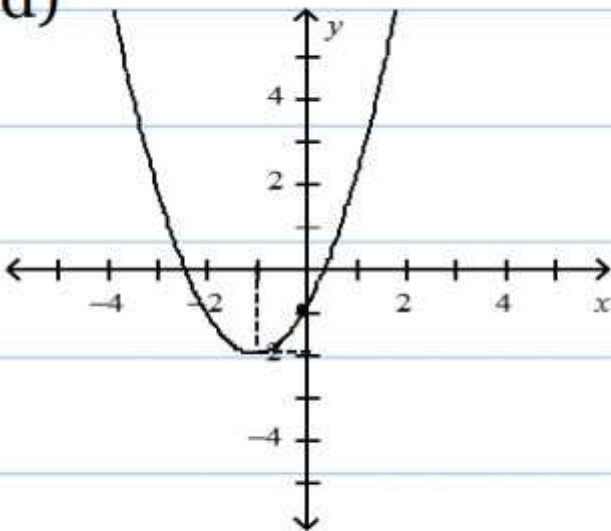
$$y = \frac{k}{x-4}$$

$$-1 = \frac{k}{3-4}$$

$$k = 1$$

$$\therefore y = \frac{1}{x-4}$$

d)



$$y = a(x+1)^2 - 2$$

$$-1 = a(0+1)^2 - 2$$

$$1 = a$$

$$\therefore y = (x+1)^2 - 2$$